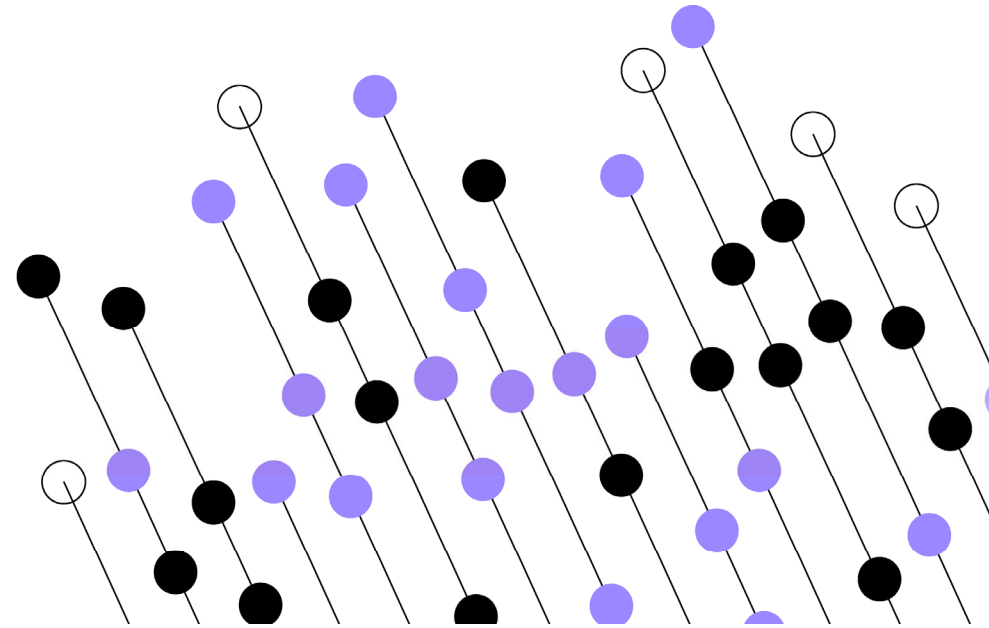


Java language basics



Variables (symbolic name for an address in memory)

Must declare a type before use

Must be initialised before use

```
public void main(String[] args) {  
    int myAge;  
  
    boolean answer = true;  
  
    String myName = "Samantha";  
  
    int i = 0, j;  
  
    myAge = 21;  
    System.out.println(i); ✓  
    System.out.println(j); ✗ // not initialised  
}
```

Pre-defined in-built primitive data types

```
byte      eightBit;  
short     sixteenBit;  
int       thirtyTwoBit;  
long      sixtyFourBit;
```

```
float      x32;  
double     x64;
```

```
Float limits    7 digits of precision  
Double limits   16 digits of precision
```

```
float f = 0.1; ❌
```

```
float f = 0.1F; ✔️
```

```
boolean isActive;  
isActive = true;
```

can only be set to true/false

java.math.BigDecimal is a **reference type** used by high-precision calculations like financial and scientific apps. Please see the annex of this chapter for details.

What is a char?

```
char    initial;    // Unicode character (16 bits)
initial = 'M';
```

```
char c = 'A';
int k = c;
System.out.println(c);

System.out.println(k);

System.out.println(c + k);
```

A	
65	ASCII code for letter 'A'
130	

Standard mathematical operators

```
int x = 4;
x = x + 5;           // x is 9

double d = 9.0;
d = d / 2;           // d is 4.5

int y = ( x % 3 );   // y is 1
```

```
int x = 9;
x = x / 2;           // x is 4!

double d = 9.0;
d = d / 2;           // d is 4.5
```

+	addition
-	subtraction
*	multiplication
/	division
%	modulus division

Compound operators

Each mathematical operator can be combined with "="

```
int x = 4, y = 6;  
x = x + y;           // x = 10
```

```
// Can be coded as  
x += y;
```

```
int x = 8;  
x *= 2;              // x = 16
```

```
int x = 8;  
System.out.println(x % 5); // displays 3 but x = 8  
  
x %= 5;              // x = 3
```

Pre- and post-fix ++ and -- operators

```
int x = 0;  
x++;           // x = 1  
++x;           // x = 2
```

```
int x = 0;  
int y = ++x;    // x=1, y=1
```

```
int x = 0;  
int y = x++;    // x=1, y=0
```

```
int x = 1;  
System.out.println(x++); // displays 1 and then x=2  
System.out.println(x);   // displays 2
```

Casting — implicit casting

```
int x = 4;
```

```
long no = x;
```



```
double d = x;
```

```
char c = 'A';
```



```
int x = c;
```

65



Casting — explicitly cast any numeric type to another type

```
double d = 4.5;  
int x = d;
```



```
double d = 4.5;  
int x = (int) d;
```

```
long lng = 5;  
int x = lng;
```



```
long lng = 5;  
int x = (int)lng;
```

```
boolean b = 1;
```



```
boolean b = true;
```

Casting strings

Must use a parse method to cast strings to numeric types.



```
String no = "123";  
int x = (int)no;
```

```
String no = "123.45";  
double d = (double)no;
```



```
int x = Integer.parseInt(no);  
double d = Double.parseDouble(no);  
float f = Float.parseFloat(no);
```



Operator precedence

Order	Operators	Comments
1	() . f(x) [] x++ x-- new	Primary
2	+ - ! ~ ++x --x (T)x	Unary
3	* / %	Multiplicative
4	+ -	Additive
5	< < > >	Bit Shift
6	< > <= >= instanceof	Relational
7	== !=	Equality

Order	Operators	Comments
8	&	Bitwise AND
9	^	Bitwise exclusive OR
10		Bitwise inclusive OR
11	&&	Logical AND
12		Logical OR
13	?:	Conditional
14	= op=	Assignment

Operator precedence example

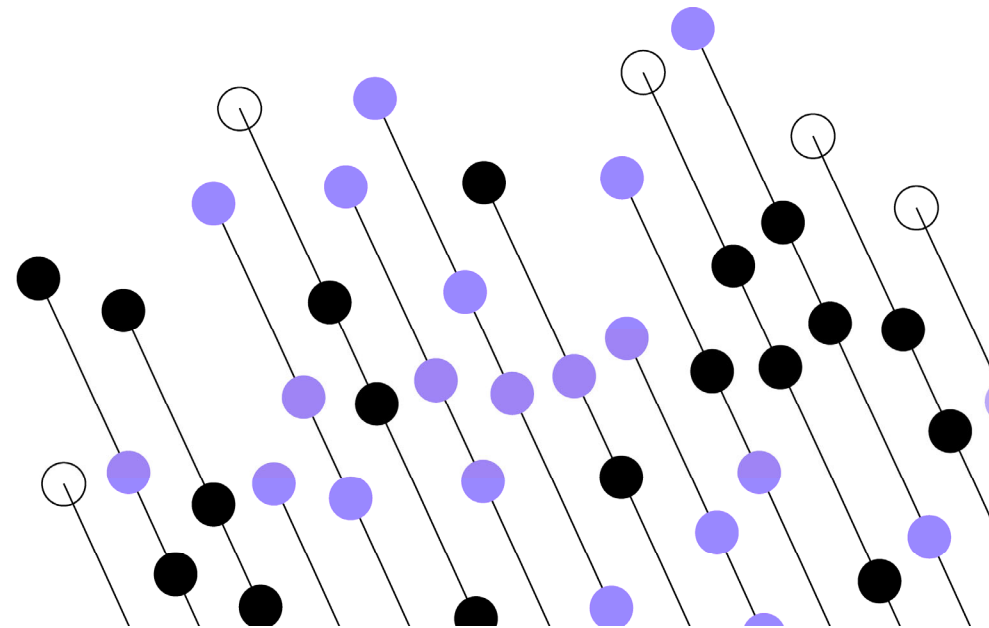
```
int p = x + (y * z);  
int q = (x + y) * z;  
bool b1, b2, b3;  
  
if (b1 && b2 || b3) {  
    // does this if b3 true?  
    // b1 and/or b2 might be true?  
}
```

What is the order or precedence?

Tip: Always use practises!



Conditionals



Introducing 'if'

```
int age = getAge();  
  
if ( age > 18 )  
{  
    // code for when over 18 years old  
}
```

```
int age = getAge();  
  
if ( age < 18 ) {  
    // code for when under 18 years old  
}  
else {  
    // code for when 18 or over  
}
```

>	greater than
>=	greater than or equal to
<	less than
<=	less than or equal to
==	equal to
!=	not equal to

Introducing 'else if'(s)

```
int mark; . . .  
  
if ( mark > 80 ){  
    System.out.println("Distinction");  
}  
else if( mark > 70 ) {  
    System.out.println("Merit");  
}  
else if( mark > 60 ) {  
    System.out.println("Pass");  
}  
else {  
    System.out.println("Try again!");  
}
```

'if' must come first

as many 'else if'(s) as needed

'else'(if needed) comes last

The ternary conditional operator (? :)

- Produces less code for simple if statements resulting in a value.
- These two examples produce the same result:

```
double salary = getSalary();  
double rate = (salary < 21000) ? 0.2 : 0.4;
```

```
double salary = getSalary();  
  
if(salary < 21000) {  
    rate = 0.2;  
}  
else {  
    rate = 0.4;  
}
```


Logical operators AND and OR

```
int var1 = 4, var2 = 2, var3 = 0;

if (var1 > var2 && var3 == 0)
{
    System.out.println( "Will we see this?" );
}
```

A

&&	AND
	OR
!	NOT

```
int var1 = 4, var2 = 6, var3 = 0;

if (var1 > var2 || var3 == 0)
{
    System.out.println( "Will we see this?" );
}
```

B

Logical operators AND and OR

```
int var1 = 1, var2 = 2, var3 = 3;
if ( var1 == 1 || var2 == 2 && var3 == 1)
{
    System.out.println( "Will we see this?" );
}
```

C



&&	AND
	OR
!	NOT

```
int var1 = 1, var2 = 2, var3 = 3;
if ( (var1 == 1 || var2 == 2) && var3 == 1)
{
    System.out.println( "Will we see this?" );
}
```

Use ()

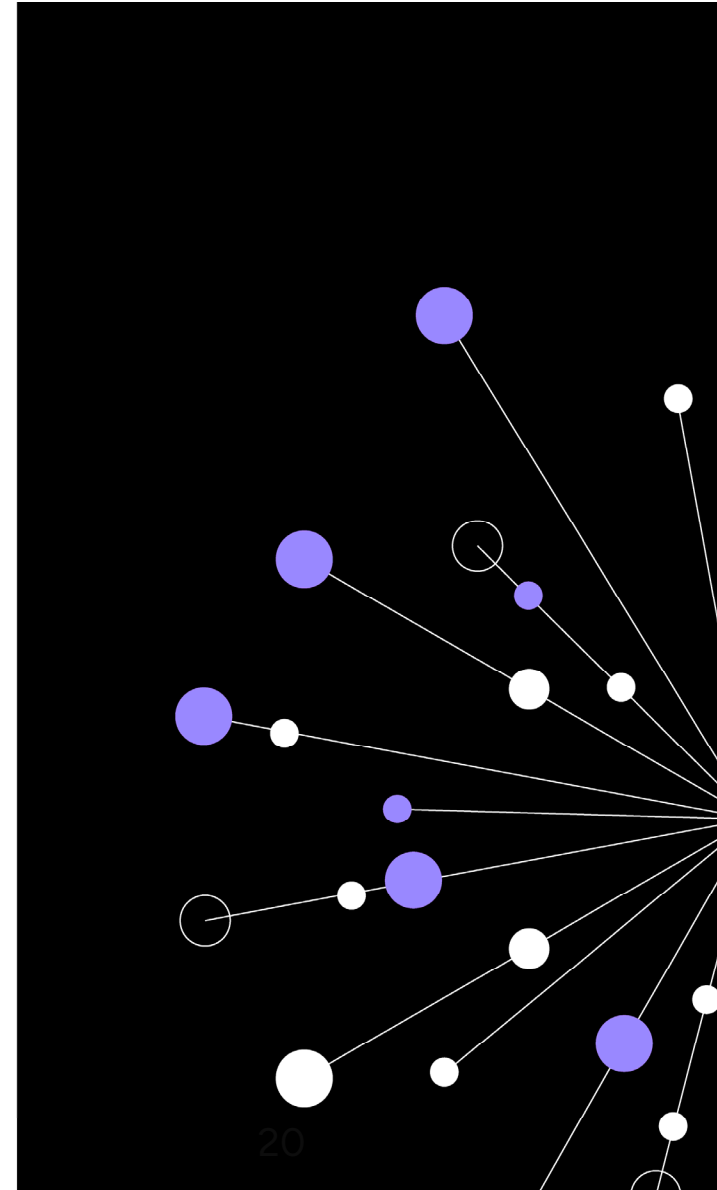
The switch statement

- Tests an integer, enum, char, or String.
- Statements may be in any order.
- Often elegant alternative to **if...else if... else**.
- Beware of accidentally dropping through to next section.
 - Compiler does not force a 'break' statement.

```
int no = 1, res;  
  
switch (no) {  
  
    case 0:  
        res = 23;  
        break;  
    case 2:  
        res = 8;  
        // break;   
    case 1: case 2:  
        res = 51;  
        break;  
  
    default:  
        res = -1;  
        break;  
  
}
```

In this chapter, we reviewed...

- Creating variables.
- Selection statements: if, else, else if, and switch.



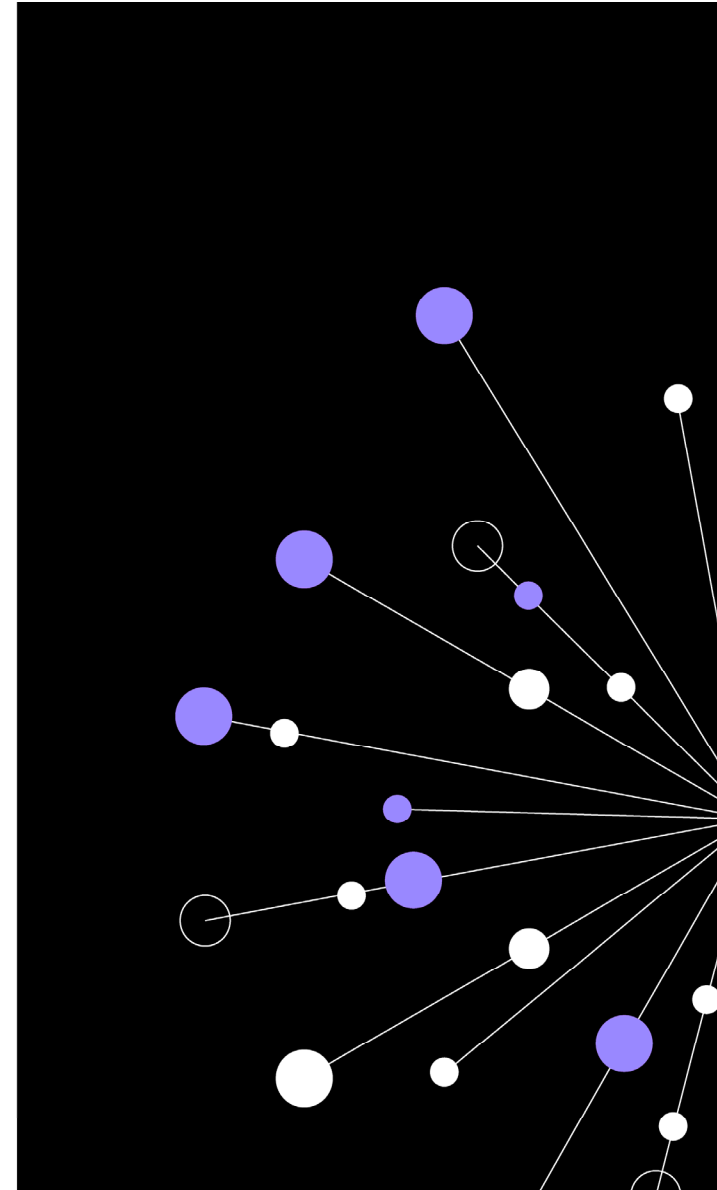
Lab

Please open the labs folder and complete the next lab.

Objectives:

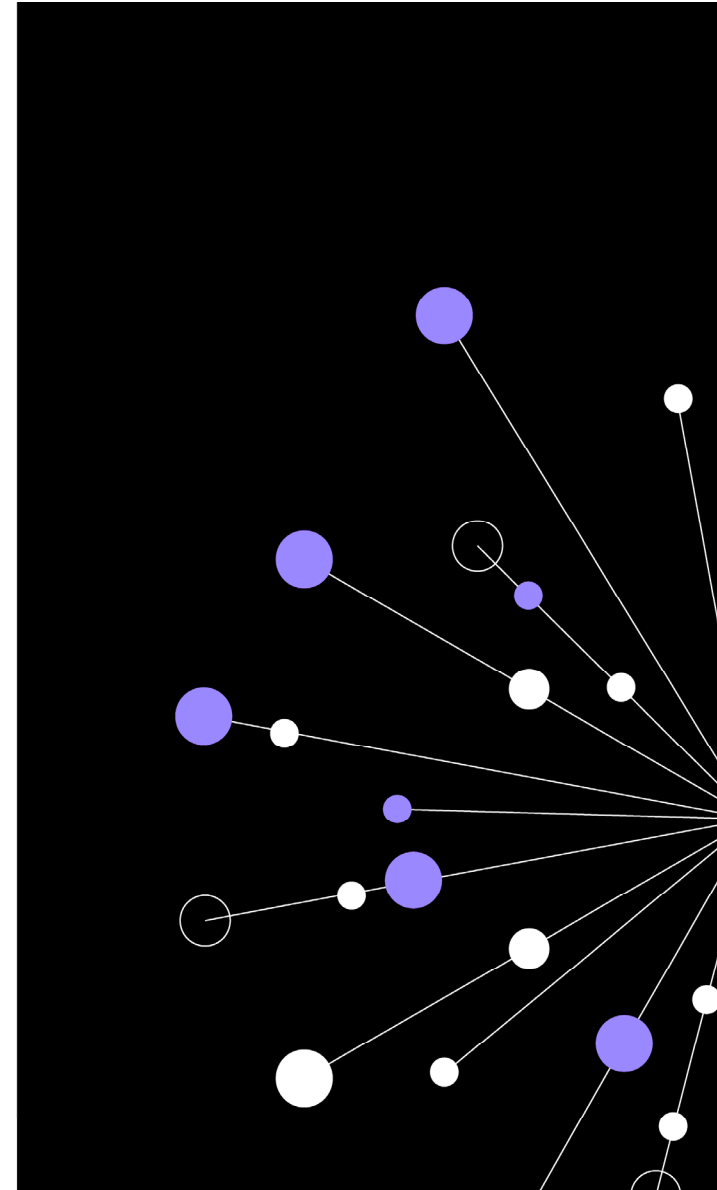
- Practice declaring and using variables.
- Convert variables from one type to another.
- Understand why some type conversions are unsuitable.
- Practice using IF and SWITCH statements.
- Use ternary type of IF statements as well as enhanced SWITCH statements.

Note: Use an IDE like Eclipse, IntelliJ, or VS Code. Avoid using the command line.



Extra material

QA



The BigDecimal for when precision matters (financial, scientific, or large-scale calculations)

```
import java.math.BigDecimal;

BigDecimal a = new BigDecimal("10.50");

BigDecimal b = new BigDecimal("2.25");

• Addition           BigDecimal sum = a.add(b);

• Subtraction         BigDecimal difference = a.subtract(b);

• Multiplication       BigDecimal product = a.multiply(b);

• Division             BigDecimal res = a.divide(b, 2, BigDecimal.ROUND_HALF_UP );
```

The BigDecimal...

```
import java.math.BigDecimal;
```

```
BigDecimal a = new BigDecimal("10.50");  
BigDecimal b = new BigDecimal("2.25");
```

- Compare

```
System.out.println(a.equals(b));           // false (because scale differs)
```

```
System.out.println(a.compareTo(b));        // +1 (because a > b) -1 if a < b and 0 if equal
```

- Cast

- double res = a.doubleValue();

The **var** keyword

Automatically infers the type of a variable at compile time.

```
var message = "Hello, Java 10!";           // Compiler infers String
var number = 42;                           // infers int
var price = 99.99;                         // infers double

for (var i = 0; i < 5; i++) {               // infers int
    System.out.println("Count: " + i);
}
```

ERROR! Where var can't be used

```
var x; // ❌ Not without assignment
```

```
class Example {  
    var name = "Not Allowed"; // ❌ Not as a field  
}
```

```
public void print(var message) { // ❌ Not as a parameter  
    System.out.println(message);  
}
```

```
public var getValue() { // ❌ Not as return type  
    return 42;  
}
```

Java 14+ enhanced switch statement with -> syntax

```
String category = "Groceries";  
double price = 0;  
price = switch (category) {  
    case "Electronics" -> 299.99;  
    case "Clothing" -> 129.99;  
    case "Groceries" -> 15.50;  
    case "Furniture" -> 2299.50;  
    default -> throw new IllegalArgumentException("Unknown category");  
};  
System.out.println("The price is £" + price);
```